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Name	Index Number	CTG
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BIOLOGY

8875/01

25th August 2011

Friday 0800-0900

Additional materials: Multiple Choice Answer Sheet

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The logo of Yishun Junior College is a shield-shaped emblem. At the top, it says "YISHUN JUNIOR COLLEGE". The shield is divided into two halves: the left half features a stylized flame or torch, and the right half features a stylized lion's head. Below the shield, there is a banner with the Latin motto "FLORESCAT CONCORDIA". The entire emblem is flanked by laurel branches.



READ THESE INSTRUCTIONS FIRST

Write in soft pencil (type B or HB is recommended).

Do not staples, paper clips, highlighters, glue or correction fluid.

Write your name, index number and CTG on the Answer Sheet in the spaces provided.

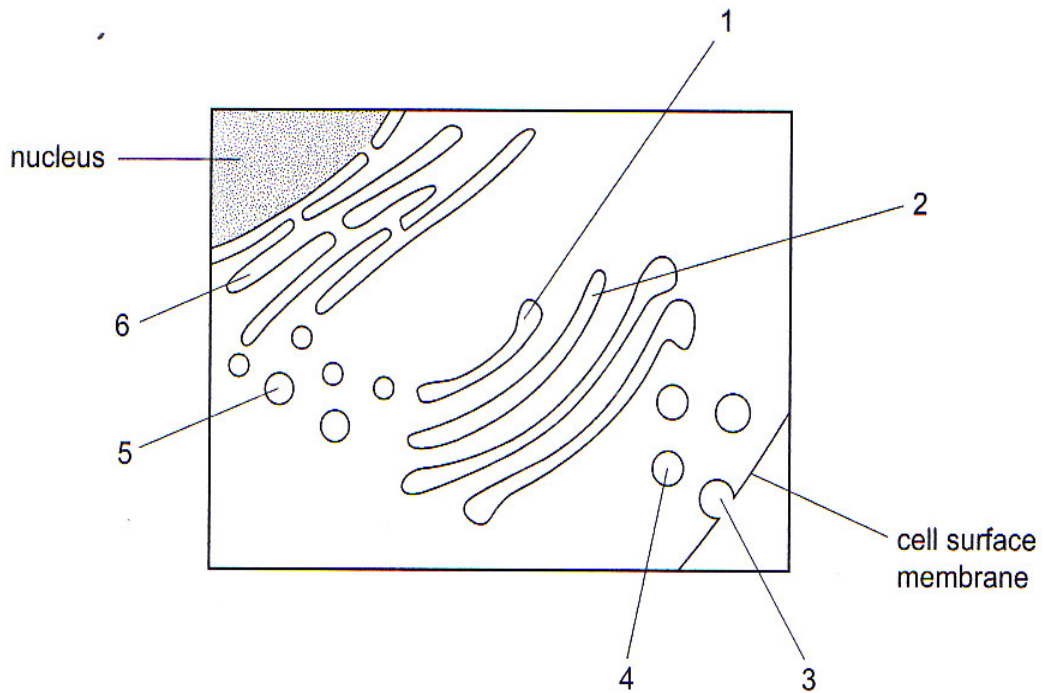
There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of **20** printed pages.

1. The diagram of an electron micrograph shows a number of membrane-bound structures within an animal cell.



What is the correct sequence of stages linking the membrane-bound structures during the production of an extracellular enzyme?

- A 2 → 1 → 3 → 4 → 5 → 6
B 3 → 4 → 2 → 1 → 5 → 6
C 6 → 5 → 1 → 2 → 4 → 3
D 6 → 5 → 4 → 3 → 1 → 2
2. Which molecule prevents the cell surface membrane from becoming too fluid or too rigid?
- A cholesterol
B glycolipid
C glycoprotein
D phospholipid

3. Diagram 1 below shows a slide micrometer with the smallest divisions 0.005mm apart and an eyepiece graticule at $\times 400$ magnification. Diagram 2 shows the same eyepiece graticule and part of a plant tissue at the same magnification.

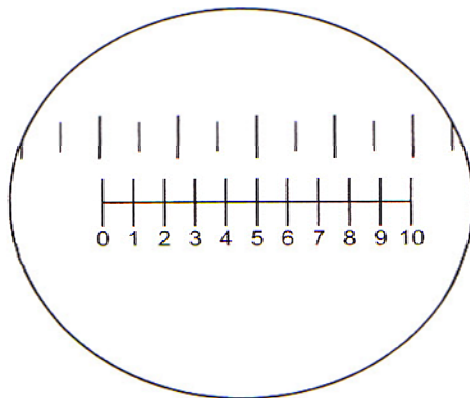


Diagram 1

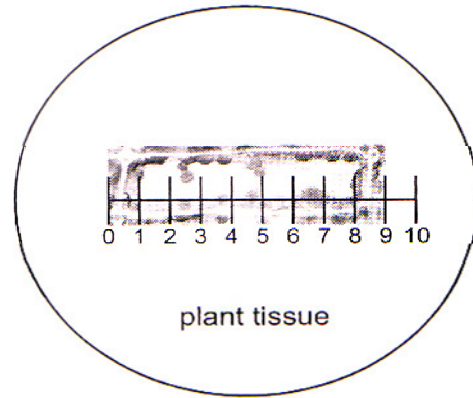


Diagram 2

What is the actual length of the part of the plant tissue in diagram 2?

- A 370nm
 - B $3.7\mu\text{m}$
 - C $37\mu\text{m}$
 - D $370\mu\text{m}$
4. Which features help to give collagen fibers their great strength?
- 1 The small size and large numbers of molecules of glycine in each polypeptide chain.
 - 2 The hydrogen bonds which hold together the three polypeptide chains in each triple helix.
 - 3 The covalent bonds which hold together adjacent triple helices.
- A 1 only
 - B 2 only
 - C 2 and 3 only
 - D 1, 2 and 3 only

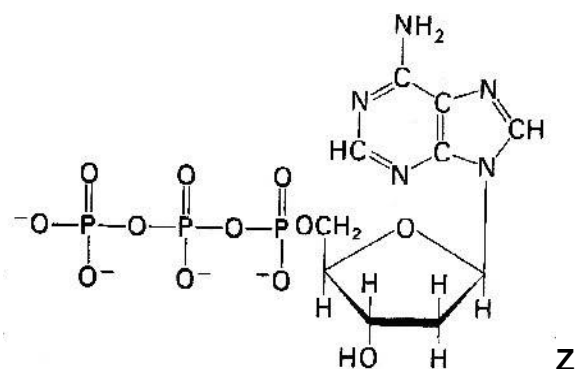
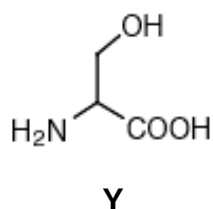
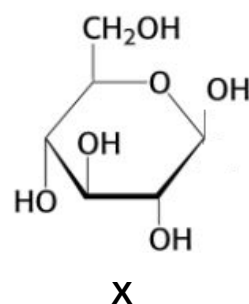
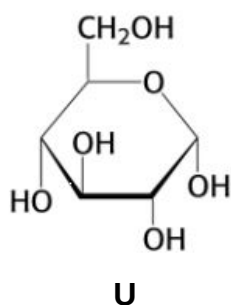
5. The correct shape of a globular protein is maintained by several types of bond.

Bond types	1	Disulfide
	2	Hydrogen
	3	Ionic
	4	Peptide

What bond types are involved in maintaining the different levels of structural complexity of a globular protein?

	Level of structural complexity		
	Primary	Secondary	Tertiary
A	4	3	1, 2, 3
B	2	1	2, 3, 4
C	2	4	2, 3, 4
D	4	2	1, 2, 3

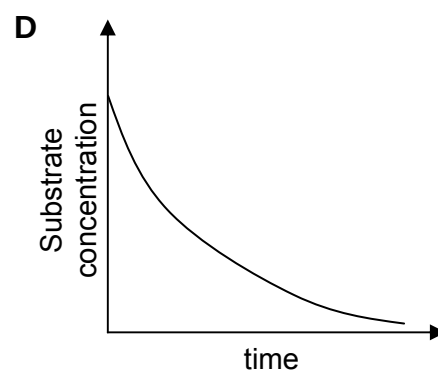
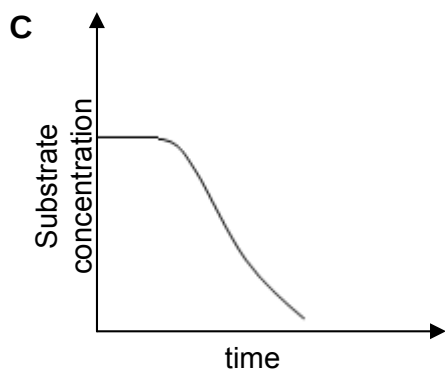
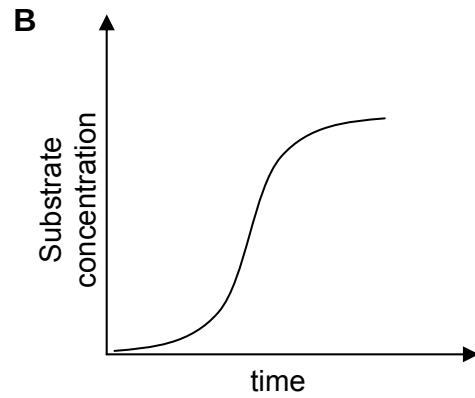
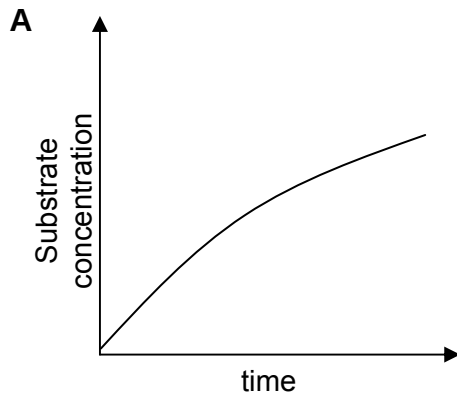
6. The diagrams below show the structures of 4 monomers.



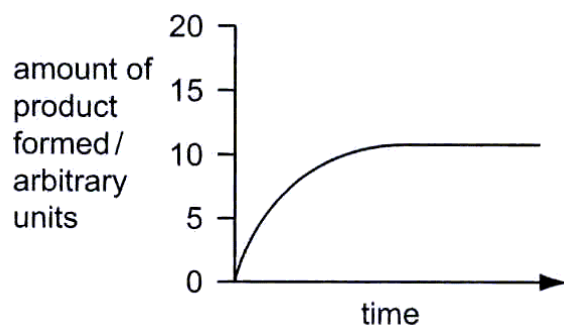
Which of the following combination of polymer, monomer and bond formed between monomers is correct?

	<i>starch</i>		<i>cellulose</i>		<i>polypeptide</i>		<i>polynucleotide</i>	
A	X	β -1,4 glycosidic bond	U	α -1,4 glycosidic bond	Z	ester linkage	Y	disulphide linkage
B	U	α -1,4 glycosidic bond	X	β -1,4 glycosidic bond	Y	peptide bond	Z	phosphoester linkage
C	Z	peptide bond	X	hydrogen bond	Z	ionic bond	U	hydrogen bond
D	X	ionic bonds	Y	peptide bond	U	hydrogen bond	Z	α -1,6 glycosidic bond

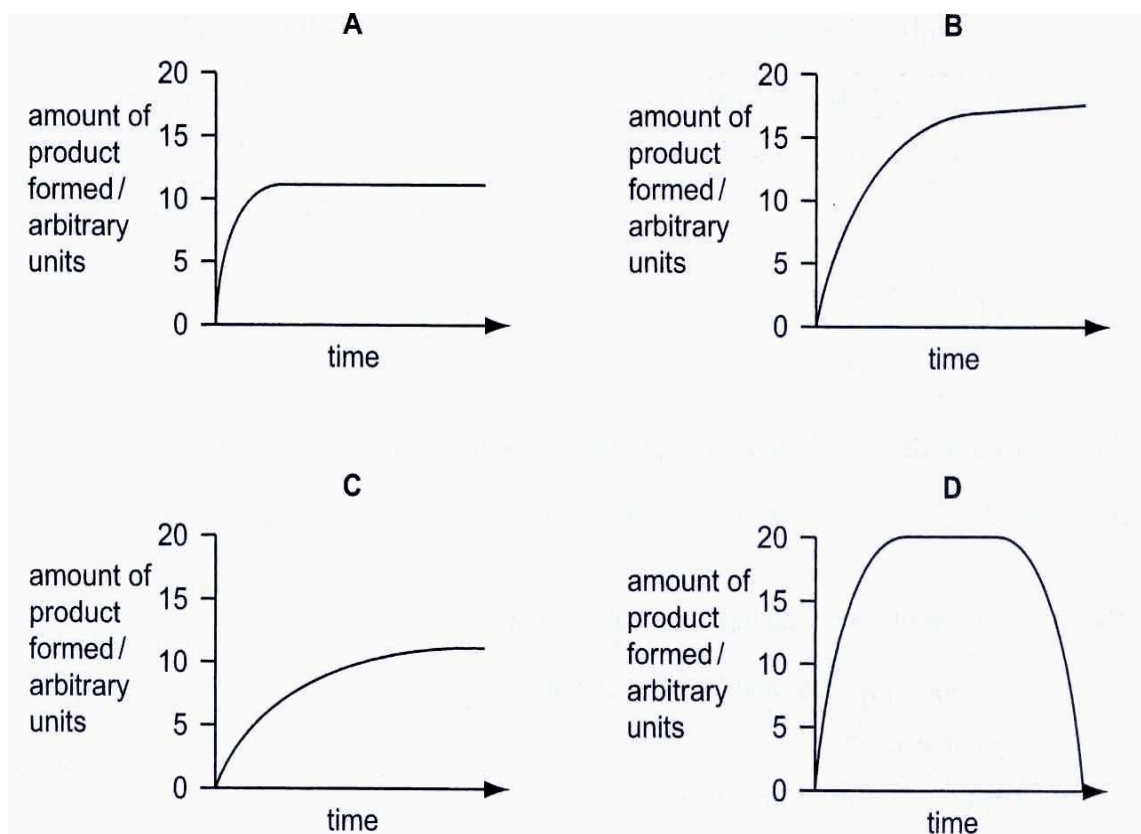
7. Which graph represents an enzyme dependent reaction?



8. The graph shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C.



Which graph shows the effect of increasing the temperature to 20°C on enzyme activity?



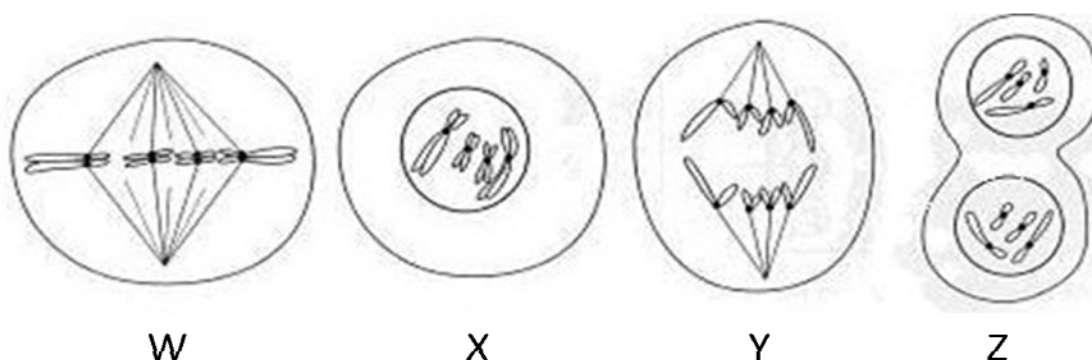
9. The diagram shows a human karyotype.



The person who has this karyotype is

- A A normal male.
- B A normal female.
- C A male with chromosomal mutation.
- D A female with chromosomal mutation.

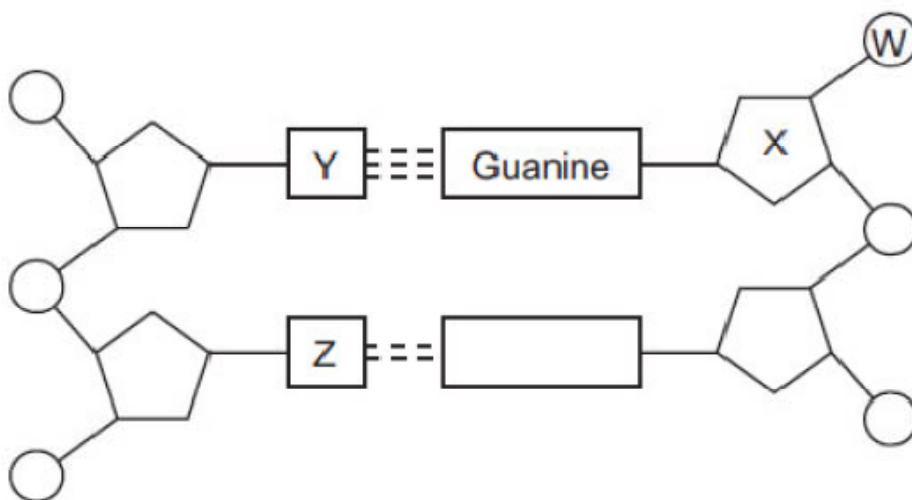
10. The diagrams show the cell cycle of a cell involved in asexual reproduction and the arrangement of two pairs of its chromosomes at different stages of the cycle.



Which combination represents the correct match between the stage of cell cycle and the arrangement of chromosomes?

	Stage of the cell cycle	Arrangements of chromosomes
A	Prophase	Z
B	Anaphase	Y
C	Metaphase	X
D	Telophase	W

11. The diagram shows part of a DNA molecule.



What does each letter represent?

	W	X	Y	Z
A	Phosphate	Deoxyribose	Guanine	Thymine
B	Phosphate	Deoxyribose	Cytosine	Thymine
C	Phosphate	Deoxyribose	Guanine	Adenine
D	Phosphate	Ribose	Cytosine	Uracil

12. Which statement about tRNA is **not** correct?

- A Hydrogen bonds between bases temporarily hold tRNA against mRNA.
- B The base sequences in the tRNA molecules are the same as the base sequence in the DNA that is being translated.
- C tRNA translates the base sequence in DNA into the amino acid sequence in a protein.
- D The specificity of a tRNA molecule depends on the specificity of the enzyme that loads it with an amino acid.

13. The diagram shows the banding pattern of two human chromosomes. **P** is a normal chromosome.



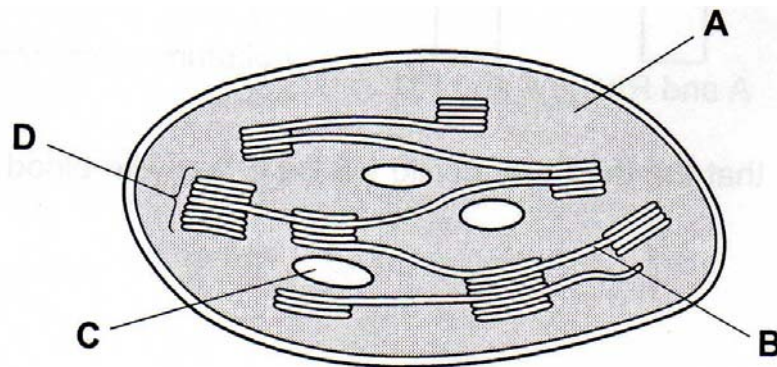
What accounts for chromosome **Q**?

- A Crossing over between sister chromatids.
 - B Inversion of part of the chromosome.
 - C Deletion of part of the chromosome.
 - D Translocation of part of another chromosome.
14. An individual **A** of unknown genotype is mated with a homozygous recessive individual **B** from the same species. All of the offspring have the same phenotype as the parent **A**. What can be concluded from this result?
- A The individual **A** is heterozygous dominant for the trait.
 - B The genotype of the individual **A** cannot be determined.
 - C The individual **A** is homozygous dominant for the trait.
 - D The environmental factors have a great influence on the phenotypic expression of this trait.
15. A woman who is a carrier for haemophilia and had phenotypically normal parents marries a man without the recessive haemophilia allele.

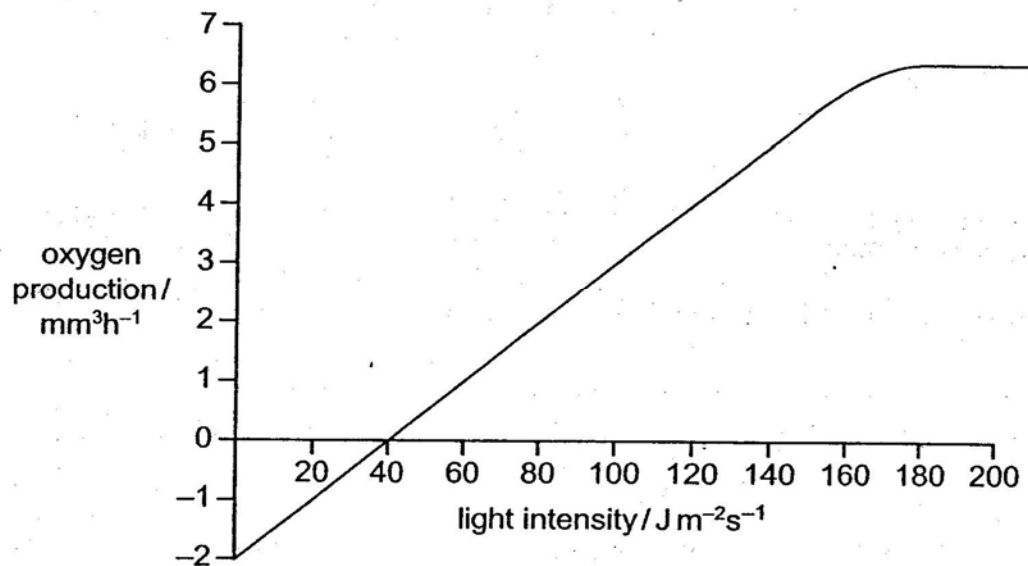
Which statement is true?

- A All of her sons will be haemophiliacs.
- B Haemophilia will not appear in any of her children.
- C Her father is a haemophiliac.
- D Her mother must have possessed the allele.

16. The diagram shows a section through a chloroplast.
Where would the products of photophosphorylation be used



17. The graph shows the relationship between oxygen production in photosynthesis and light intensity for a unicellular green organism in 0.02% sodium hydrogencarbonate solution.



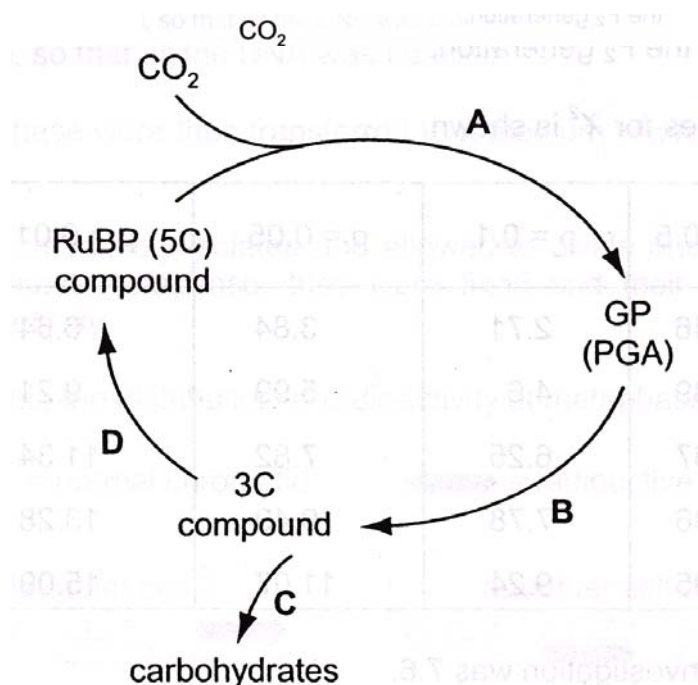
What is the most likely explanation for the graph levelling off at 180 Jm⁻²s⁻¹?

- A Light is limited and carbon dioxide is saturated.
- B Light is limited and the temperature is below optimum.
- C Light is saturated and carbon dioxide is limited.
- D Light is saturated and the temperature is above optimum.

18. Which combination correctly describes the light-dependent reactions of photosynthesis?

	Site of reaction in chloroplast	ATP	NADP
A	Stroma	$\text{ATP} \rightarrow \text{ADP} + \text{P}_i$	oxidised
B	Granum	$\text{ATP} \rightarrow \text{ADP} + \text{P}_i$	reduced
C	Stroma	$\text{ADP} + \text{P}_i \rightarrow \text{ATP}$	oxidised
D	Granum	$\text{ADP} + \text{P}_i \rightarrow \text{ATP}$	reduced

19. Which step in the Calvin cycle is a reduction reaction requiring reduced NADP?

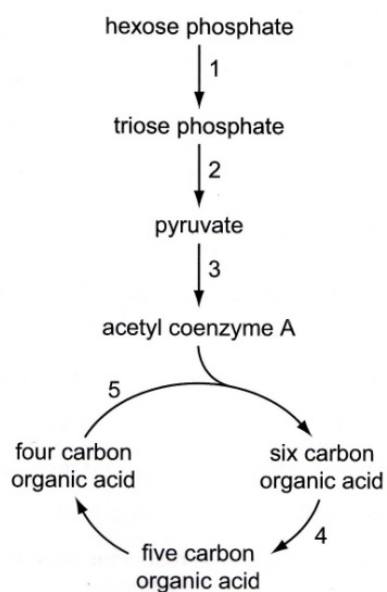


20. One reduced NAD molecule yields about 2.5 molecules of ATP and one reduced FAD molecule yields about 1.5 molecule of ATP.

What are the correct yields of reduced NAD, reduced FAD and ATP during respiration of one glucose molecule?

	glycolysis	Link reaction	Krebs cycle	Oxidative phosphorylation
A	2 reduced NAD 2 ATP	2 reduced NAD 2 ATP	6 reduced NAD 2 reduced FAD	28 ATP
B	2 reduced FAD 2 ATP	2 reduced FAD 2 ATP	6 reduced NAD 2 reduced FAD	24 ATP
C	2 reduced FAD 2 ATP	2 reduced FAD	6 reduced FAD 2 reduced NAD 4 ATP	24 ATP
D	2 reduced NAD 2 ATP	2 reduced NAD	6 reduced NAD 2 reduced FAD 2 ATP	28 ATP

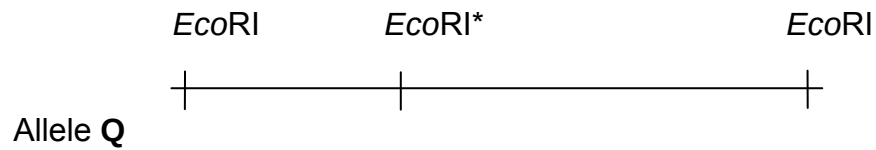
21. The flow diagram shows some of the stages in respiration.



During which stages do oxidative decarboxylation occur?

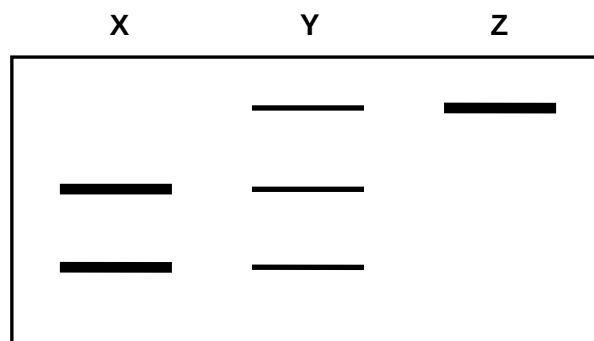
- A 1 and 3
 - B 2 and 3
 - C 3 and 4
 - D 4 and 5
22. Which process releases ATP during anaerobic respiration?
- A Carboxylation
 - B Oxidative phosphorylation
 - C Reduction of pyruvate
 - D Substrate level phosphorylation

23. The following diagram shows the recognition site of restriction enzyme *EcoRI* on a normal allele.



* indicates the polymorphic site which is missing in the recessive allele **q**.

DNA fragments from three different individuals, **X**, **Y** and **Z**, were subjected to restriction digestion by *EcoRI* and separated by gel electrophoresis. The results are shown below.



Deduce the genotype of individual **Z**.

- A QQ
- B qq
- C Qq
- D X^QY

- 24 Which of the following statement about stem cells is correct?
- A Stem cells are specialised cells that have yet to express the genes and proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
 - B Stem cells are fully differentiated cells that reside under the surface of epithelia, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
 - C Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter that remains as a stem cell, and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.
 - D Stem cells are undifferentiated embryonic cells that persist in the adult, and can give rise to all the cell types in the body.

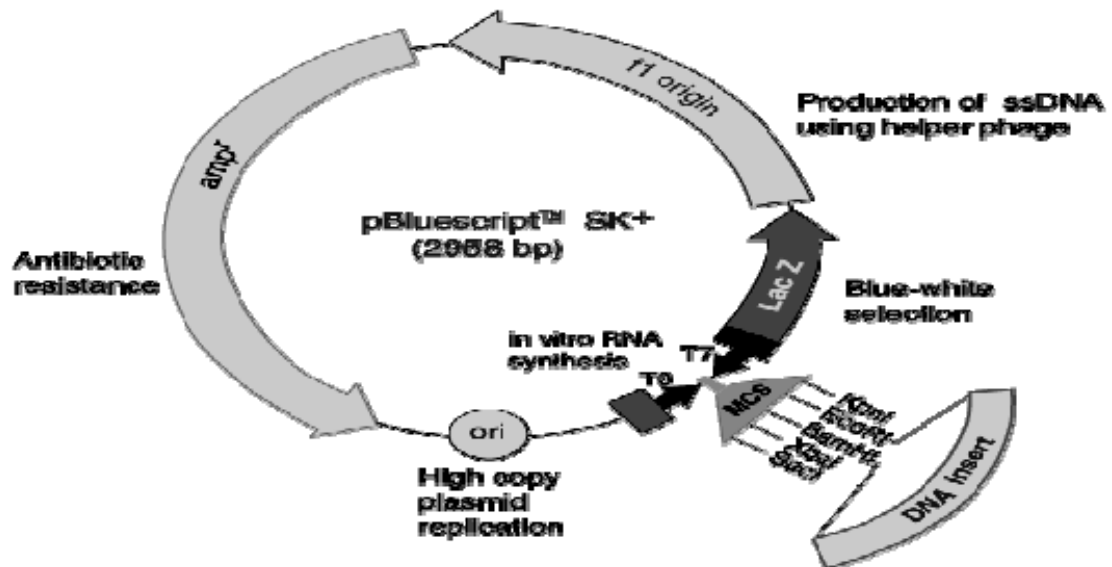
25. A researcher wishes to clone and express a gene. In the following gene sequence, N_x , denotes an unspecified sequence.

5' – AATCGCCGAGCTCGGTTACG(N_x)TAGGCATCTCGAGCCTAGG – 3'

The restriction endonucleases to be used for this project are *SacI* and *XhoI*. The recognition site for *XhoI* is CTCGAG. What is the recognition site for *SacI*?

- | | |
|----------|----------|
| A GAGCTG | B ATCGCA |
| C GGCCGA | D GCCGAG |
- 26 The completion of the Human Genome Project revealed that the human genome contains fewer genes than expected, not significantly more than simpler organisms. Which of the following statements could explain this phenomenon?
- 1 RNA transcripts of human genes are more likely to undergo alternative splicing.
 - 2 Polypeptide domains are combined in a variety of ways.
 - 3 Post-translational processing adds diversity to the resulting polypeptides.
 - 4 Gene expression patterns in humans are often more complex than those in other organisms.
- A 1 only
 - B 1 and 2 only
 - C 2, 3 and 4
 - D 1, 2, 3 and 4

- 27 The following diagram shows a plasmid which is used for genetic engineering purposes.



A eukaryotic gene is inserted in the plasmid. Which of the following statements are true about bacteria cells transformed with this recombinant plasmid?

- 1 Survive on ampicillin-containing medium
 - 2 X-gal is hydrolysed
 - 3 Multiple copies of the same plasmid in the cell
 - 4 One copy of the plasmid in the cell
- A 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 4

28 Which one of the following statements about natural selection is **incorrect**?

Natural selection may

- A occur in stable environmental conditions.
- B cause new genes to appear in a gene pool.
- C result in an increase in the frequency of favourable genes in a gene pool.
- D result in an increase in the percentage of well-adapted individuals in a population.

29 DNA-DNA hybridisation is a technique used to determine the similarity of DNA from different species.

In this technique:

- DNA is extracted from different species and cut into pieces using special enzymes;
- the DNA is heated to separate the double strands into single strands;
- single strands of DNA from different species are mixed together and cooled to form double strands of hybrid DNA.

The following table shows the temperature (in °C) needed to separate the hybrid DNA strands of four different species, **Q**, **R**, **S**, and **T**:

	Q	R	S	T
Q	86	85	81	82
R		87	82	81
S			86	85
T				87

Which one of the following statements is supported by the data provided?

- A **Q** is more closely related to **R** than **S** is to **T**.
- B **Q** is more closely related to **S** than it is to **R** or **T**.
- C **S** is more closely related to **R** than **Q** is to **T**.
- D **S** is more closely related to **T** than it is to **R** or **Q**.

- 30** The α , β and γ globin chains of human, chimpanzee, gorilla and gibbon haemoglobin were analysed. The number of differences in the amino acid sequences in comparison with the human molecules is shown in the table below.

Species	Number of differences in amino acid sequence from human molecules		
	α globin	β globin	γ globin
Chimpanzee	3	3	5
Gorilla	4	5	4
Gibbon	6	6	5

What do the differences suggest?

- A** Humans and gorillas share a common ancestor more recently than humans and gibbons.
- B** Humans and gibbons share a common ancestor more recently than humans and chimpanzees.
- C** Humans evolved from chimpanzees.
- D** Humans and gorillas do not share a common ancestor.

--End of Paper --